

NORWEGIAN FJORDS

Sheltered havens of a convoluted coast

The coast of Norway is dominated by steep-sided valleys gouged out by glaciers over several ice ages. These were then flooded by the sea, forming long, narrow inlets with vertiginous rock walls known as fjords. The waters are fully marine, but sheltered, largely inaccessible from land, and often extremely deep. Despite their high latitude, the fjords usually remain ice-free all year thanks to the warming influence of the Gulf Stream. They support an abundance of resident and migratory fish, seals and porpoises, and sea birds, and the world's largest accumulation of deep-water coral reefs.

Cold-water corals

Deep-water corals were first discovered in 1869, but it took more than a century for their size and extent to be revealed. The main reef-forming coral in the Atlantic is *Lophelia pertusa*, and lophelia reefs more than 8 miles

(13 km) long and 100 ft (30 m) high have been found off the coast of Norway. Some of the reefs are thousands of years old. The reefs in the more shallow, but still cold waters of the fjords were discovered in 2000. They greatly enhance the ecological value of the fjords, giving shelter to a wide range of marine invertebrates and providing nursery and feeding areas for fish.

The mountainous land adjacent to and between the fjords is cloaked in coniferous and deciduous forests, dotted with glacial lakes and summer-grazing pastures in high valleys. Above 5,600 ft (1,700 m), the trees and meadows give way to alpine plants and snowy peaks.

BOOM AND BUST
Norway lemmings can breed very quickly. When good conditions prevail, populations boom, leading to large-scale dispersal and the myth that lemmings hurl themselves into fjords to escape overcrowding. Populations crash due to limited food.

NORWAY LEMMING

ATLANTIC HERRING

SPAWNING GROUNDS
The Atlantic herring migrates to Norwegian fjords to spawn when it reaches five years. It is a key food item for several other fish, seals, cetaceans, and sea birds, and so all these animals suffer from overfishing of herring stocks by humans.

WHITE-BACKED WOODPECKER

Sognefjord is the longest ice-free fjord in the world

The largest lophelia reefs are more

HAMMERED HOME

The relatively undisturbed coastal forests of Norway support dense populations of the white-backed woodpecker. Considered an indicator of healthy and mature ecosystems, its excavations provide nest holes for many other species.

